

Chapter 4.

4.1.

$$(a) Y_{eff} = (1 + \frac{0.1}{2})^2 - 1 = 0.1025$$

$$(b) Y_{eff} = (1 + \frac{0.1}{4})^4 - 1 \approx 0.1038$$

$$(c) Y_{eff} = e^{0.1} - 1 \approx 0.1052$$

$$4.2. (e^r)^t = 2 \Rightarrow e^{at} = 2$$

$$\Rightarrow t = \frac{\log 2}{0.1} \approx 6.93 \text{ years}$$

$$4.3. (1+0.5)^t = 4 \Rightarrow (1+0.5)^t = 2^2$$

By the double rule, we have

$$t = 2 \left(\frac{0.7}{r} \right) = 2 \cdot \frac{0.7}{0.05} = 28 \text{ years}$$

If $r = 0.04$, we then have

$$t = 2 \cdot \left(\frac{0.7}{0.04} \right) = 35 \text{ years}$$

$$4.4. (1+r)^n = 3$$

$$n = \frac{\log 3}{\log(1+r)} \approx \frac{1.1}{r}$$

$$4.5. A((1+r) + (1+r)^2 + \dots + (1+r)^{60}) = 1000000$$

$$\text{where } r = \frac{0.06}{12} = 0.005$$

thus

$$A \sum_{i=1}^{60} 1.005^i = 1000000$$

$$A \frac{1.005 - 1.005^{61}}{1 - 1.005} = 1000000$$

$$A = \frac{1000000(-0.005)}{1.005(1 - 1.005^{60})}$$

$$\approx 1426.15$$

$$4.6. PV = -\frac{1000}{1.06} - \frac{1200}{1.06^2} + \frac{800}{1.06^3} + \frac{900}{1.06^4} + \frac{800}{1.06^5}$$

$$\approx -29.01 < 0$$

not a worthwhile investment.

$$4.7. S_1 = \frac{20}{1+r} + \frac{20}{(1+r)^2} + \frac{20}{(1+r)^3} + \frac{15}{(1+r)^4}$$

$$+ \frac{10}{(1+r)^5} + \frac{5}{(1+r)^6}$$

$$S_2 = \frac{10}{1+r} + \frac{10}{(1+r)^2} + \frac{15}{(1+r)^3} + \frac{20}{(1+r)^4}$$

$$+ \frac{20}{(1+r)^5} + \frac{20}{(1+r)^6}$$

$$(a) r = 3\% \Rightarrow S_1 = 82.71, S_2 = 84.63$$

$\Rightarrow S_2$ is preferable.

$$(b) r = 5\% \Rightarrow S_1 = 78.37, S_2 = 78.60$$

$\Rightarrow S_2$ is preferable

$$(c) r = 10\% \Rightarrow S_1 = 69.01, S_2 = 65.99$$

$\Rightarrow S_1$ is preferable.

$$4.8. PV = -10,000 + \sum_{i=1}^{10} \frac{500}{(1+r_2)^i} + \frac{10000}{(1+r_2)^{10}}$$

$$(a) r = 6\% \Rightarrow Y_e = (1 + 1/12)^6 - 1 \approx 0.03038 \Rightarrow PV \approx 1670.62$$

$$(b) r = 10\% \Rightarrow Y_e \approx 0.05105 \Rightarrow PV \approx -80.9196$$

$$(c) r = 12\% \Rightarrow Y_e \approx 0.6152 \Rightarrow PV \approx -841.82$$

$$4.9. 1000 + 160 \left(\frac{1}{1+r} + \frac{1}{(1+r)^2} + \dots + \frac{1}{(1+r)^{24}} \right) = 4200$$

$$160 \sum_{i=1}^{24} \frac{1}{(1+r)^i} = 3200$$

$$\frac{1}{1+r} \frac{1 - \frac{1}{(1+r)^{24}}}{1 - \frac{1}{1+r}} = 20$$

$$\frac{1 - \frac{1}{(1+r)^{24}}}{r} = 20 \Rightarrow 1 - \frac{1}{(1+r)^{24}} - 20r = 0$$

$$r \approx 0.0152$$

4.11 The cost flow sequence (in units of \$1000.

Buy at beginning of year 1: 22, 7, 8, 9, 10, -4

--- year 2: 9, 25, 7, 8, 9, -9

year 3: 9, 11, 28, 7, 8, -14

year 4: 9, 11, 13, 31, 7, -19

Interest rate 10% gives us

$$PV_1 = 22 + \frac{7}{1.1} + \frac{8}{1.1^2} + \frac{9}{1.1^3} + \frac{10}{1.1^4} - \frac{4}{1.1^5} = 46.08$$

$$PV_2 = 44.08, PV_3 = 44.17, PV_4 = 46.02$$

The company should buy a new machine one year from now.

4.13. $PV = 10,000 + 10,000(e^{-r})^{10}$ ^{compounded continuously}

a) $r = 0.02 \Rightarrow PV = 18,187.31 > 16,000$
which is not preferable

b) $r = 0.05 \Rightarrow PV = 16,065.31 > 16,000$
which is not preferable.

c) $r = 0.1 \Rightarrow PV = 13,678.79 < 16,000$
which is preferable.

4.14. The cash flow sequence is

time (yrs)	0	0.5	1	...	4.5	5
Cash	-1000	30	30		30	1030

with a continuously compounded interest 5%.

$$PV = -1000 + \sum_{i=1}^9 \frac{30}{(e^{0.05/2})^i} + \frac{1030}{(e^{0.05/2})^{10}}$$

$$\approx 40.94.$$

4.15. The amount of interest earned after n days is

$$I = 100(e^{(0.06/365)n} - 1)$$

$$n = 30, I \approx 0.4944$$

$$n = 60, I \approx 0.9912$$

$$n = 120, I \approx 1.9922$$

4.17. You have to pay after three years

$$1000e^{3r} + 2000e^{2r} + 3000e^r$$

4.18. You would pay the present value of payments

$$\frac{3}{1.05} + \frac{5}{1.05^2} + \frac{6}{1.05^3} + \frac{5}{1.05^4} = 6.744$$

4.19. $20 + \frac{10}{1+r} > \frac{34}{1+r}$
 $20(1+r) > 24$
 $r > 0.2$

4.20

$$104e^{-r} = 110e^{-2r}$$

$$e^r = \frac{110}{104}$$

$$r = \log\left(\frac{110}{104}\right) \approx 0.0561.$$

4.23. b: 100, 140, 131

c: 90, 160, 120

$$B_1 = 100, B_2 = 240, B_3 = 371$$

$$C_1 = 90, C_2 = 250, C_3 = 370$$

$$B_1 = 100 > 90 = C_1$$

$$B_1 + B_2 = 340 = 340 = C_1 + C_2$$

$$B_1 + B_2 + B_3 = 711 > 710 = C_1 + C_2 + C_3$$

Thus by Proposition 4.21, the cash flow b is preferable for any positive interest rate.

4.24. (a) $\frac{110}{(1+r)^2} = 100 \Rightarrow r = \sqrt{\frac{110}{100}} - 1$
 ≈ 0.0488

(b) Let R be the rate of return.

$$\frac{120}{(1+R)^2} = 100 \Rightarrow R = \sqrt{1.2} - 1$$

$$\frac{100}{(1+R)^2} = 100 \Rightarrow R = 0$$

$$\Rightarrow E(R) = \frac{1}{2}(\sqrt{1.2} - 1 + 0)$$

 $\approx 0.0477.$

4.35 $1000 \cdot (e^{-0.08})^{10} = 1000 \cdot e^{-0.8}$
 $\approx 449.33.$

4.26. $100 = \frac{40}{1+r} + \frac{70}{(1+r)^2}$

$$100r^2 + 160r - 10 = 0$$

$$\Rightarrow r = \frac{-160 + \sqrt{160^2 + 4 \cdot 100 \cdot 10}}{2 \cdot 100}$$

$$\approx 0.06023$$

$$4.28. P\left(\frac{X_1}{1.1} + \frac{X_2}{1.12} > 100\right) = P(1.1X_1 + X_2 > 121)$$

$1.1X_1 + X_2$ is a normal random variable
with mean $\mu = 1.1 \times 60 + 60 = 126$ and

$$\text{variance } \sigma^2 = 1.1^2 \times 25 + 25 = 55.25.$$

Thus

$$P(1.1X_1 + X_2 > 121) = P\left(Z > \frac{121 - 126}{\sqrt{55.25}}\right)$$

$$\approx P(Z < 0.6727)$$

$$= \Phi(0.6727)$$

$$= 0.7494$$

$$4.31. -1000 \times 1.08 = -1080$$

$$(-1080 + 900) \times 1.08 = -194.4$$

$$(-194.4 + 800) \times 1.05 = 635.88$$

$$(635.88 - 1200) \times 1.08 = -609.336$$

$$-609.336 + 700 = 90.664 > 0$$

So you should invest.

$$4.32. \frac{d\bar{Y}(t)}{dt} = \frac{d}{dt} \left(\frac{1}{t} \int_0^t Y(s) ds \right)$$

$$= \frac{Y(t)}{t} - \frac{1}{t^2} \int_0^t Y(s) ds$$

$$= \frac{tY(t) - \int_0^t Y(s) ds}{t^2}$$

If $Y(t)$ is a nondecreasing function
of t , then $Y(s) \leq Y(t)$ for $s \leq t$.

$$\Rightarrow \int_0^t Y(s) ds \leq t \cdot Y(t).$$

$$\Rightarrow \frac{d\bar{Y}(t)}{dt} \geq 0$$

$\Rightarrow \bar{Y}(t)$ is a nondecreasing function of t .

$$4.33. \text{Proof: } P(t) = e^{-\int_0^t Y(s) ds}$$

$$= e^{-t\bar{Y}(t)}$$

$$\text{thus } P(\alpha t) = e^{-\alpha t \bar{Y}(\alpha t)}$$

$$(P(t))^\alpha = e^{-\alpha t \bar{Y}(t)}$$

Since $\bar{Y}(\alpha t) \leq \bar{Y}(t)$ for $0 < \alpha \leq 1$, we get

$$P(\alpha t) \geq (P(t))^\alpha.$$